

TA211 PROJECT REPORT



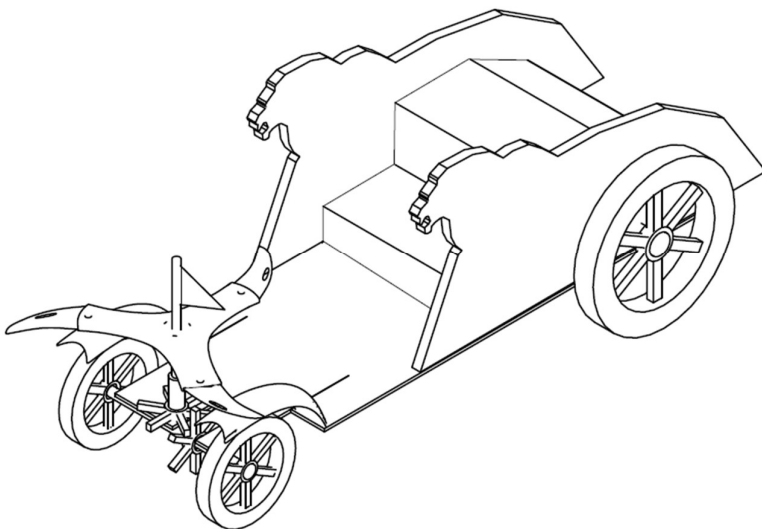
INDIAN INSTITUTE OF TECHNOLOGY KANPUR

COURSE INSTRUCTOR – Prof. Shashank Shekhar

LAB IN-CHARGE – Mr. Rakesh Dixit

TUTOR - Prof. Shashank Shekhar

CHARIOT



Group No – 9

(Monday)

- 240589 - LAKSHYA ARUKIYA
- 240627 - MANNAT SHRIVASTAV
- 240630 - MANTHAN J
- 240639 - MAYANK RAJ
- 240640 – MAYANK VERMA
- 240646 – MEENAKSHI MEENA

TABLE OF CONTENT

CONTENTS	PAGE NO.
INTRODUCTION	3
MOTIVATION	4
ACKNOWLEDGEMENT	5
WORK DISTRIBUTION	6
MATERIALS LIST	7
ISOMETRIC ASSEMBLY	8
ORTHOGRAPHIC ASSEMBLY	9
BASE	10
SIDE VIEW	11
FAN	12
FLAG	13
FRONT TYRES	14
SEAT	15
FRONT TYRE GUARD	16
REAR TYRES	17
GEARS	18

1 Introduction

Our project is named "Bhallaldev's Chariot", inspired by the grand war chariot shown in the movie Baahubali. This project was undertaken as part of the TA211 course to apply and integrate the various manufacturing processes we learned in the lab into a single working prototype.

The model consists of a base frame, side plates, seat, wheels, axles, gears, and a front rotating chopper mechanism, all fabricated using mild steel and GI sheets. A wide range of manufacturing and fabrication techniques have been employed in making this chariot, including:

- **Sheet Metal Forming** for base, side plates, guards, and seat
- **Welding & Brazing** for joining structural components and supporting parts
- **Cutting & Profile Cutting** for gears, fan blades, and decorative features
- **Drilling & Turning** for axles, wheel hubs, and shaft alignments
- **Moulding, Casting & Forging** (demonstration-based) for shaping and conceptualizing parts like wheels and joints
- **Grinding & Finishing** to achieve accuracy and smoothness in assembled parts
- **Plasma Cutting** for precise profile designs and shapes

1.1 Key Features

The key highlight of the model is the front chopper mechanism, which is powered through a combination of horizontal and vertical gears. When the knob is pushed forward, the motion is transferred to the gears, which in turn rotate the chopper. This demonstrates the principle of power transmission and motion conversion, where linear motion is effectively translated into rotational motion.

By bringing together processes such as fabrication, machining, joining, and forming, the project not only recreates a cinematic design but also serves as a practical application of manufacturing engineering principles. It reflects how creativity, design thinking, and technical knowledge can be integrated to construct a functional mechanical prototype.

The chariot features distinctive elements including decorative side plates, functional wheels with proper axle systems, a comfortable seating arrangement, and the innovative gear-driven chopper mechanism that serves as the centerpiece of the design.

2 Motivation and Project Objectives

2.1 Motivation

The idea for our project was inspired by the movie Baahubali, where Bhallaldev's war chariot was depicted as a powerful and innovative design. We were intrigued by its unique structure and mechanism, and decided to recreate a scaled version by applying the manufacturing techniques learned in TA211.

This project motivated us to bring together a variety of fabrication processes such as sheet metal forming, welding, plasma cutting, drilling, grinding, brazing, moulding, casting, and forging into a single integrated model. By designing and fabricating a gear-based rotating chopper mechanism, we were able to explore practical concepts of mechanical linkages, power transmission, and motion conversion.

The challenge of recreating a cinematic design using real-world manufacturing processes pushed us to think creatively about material selection, process optimization, and design modifications. We had to consider factors such as structural integrity, functionality, and aesthetic appeal while working within the constraints of available materials and equipment.

Working on this project fostered creativity, teamwork, and problem-solving skills, while also reinforcing our theoretical understanding of manufacturing processes. It encouraged us to think beyond conventional lab exercises and apply our knowledge in an innovative way.

The project also provided an opportunity to understand the importance of planning, coordination, and quality control in manufacturing. Each team member contributed their unique skills and perspectives, making the collaborative effort both challenging and rewarding.

Furthermore, the hands-on experience gained through this project helped bridge the gap between theoretical knowledge and practical application. We encountered real-world challenges such as material limitations, dimensional tolerances, and assembly complexities that are not always apparent in textbook examples.

Overall, this project served as a valuable learning experience, strengthening both our technical skills and our ability to integrate engineering concepts with creative design. It demonstrated how entertainment and popular culture can serve as inspiration for meaningful engineering projects.

2.2 Project Objectives

- To integrate multiple manufacturing processes learned in TA211 into a single comprehensive project

- To demonstrate practical application of mechanical design principles through a functional prototype
- To create a working model with moving parts and gear mechanisms that showcase power transmission concepts
- To explore motion conversion principles by transforming linear motion into rotational motion
- To develop teamwork, project management, and collaborative problem-solving skills
- To showcase creativity in engineering design while maintaining structural and functional integrity
- To gain hands-on experience with various fabrication techniques and manufacturing equipment
- To understand the challenges and considerations involved in translating conceptual designs into physical prototypes

3 Acknowledgement

We would like to express our deepest gratitude to Prof. Shashank Shekhar for his constant guidance, encouragement, and valuable feedback throughout the course of this project. His insights into manufacturing processes and design thinking greatly enriched our learning experience.

We are also thankful to the TA211 teaching staff and laboratory instructors, whose technical expertise and support helped us successfully carry out different fabrication processes such as welding, brazing, moulding, casting, and machining.

Our sincere appreciation goes to the workshop staff for their assistance in operating machines and providing the necessary tools and materials.

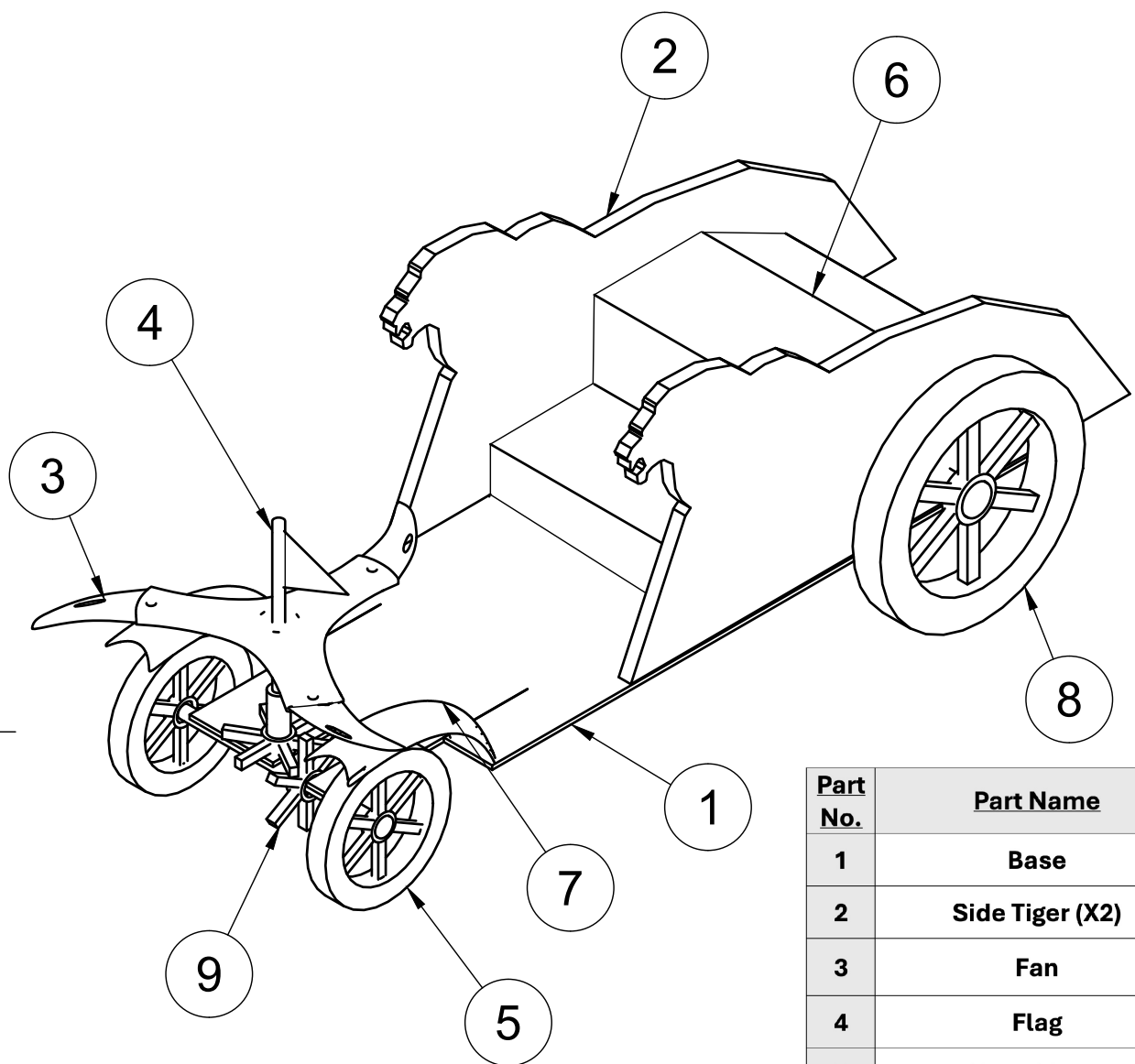
Finally, we thank all our team members for their collaborative effort, dedication, and creativity, without which this project would not have been possible.

MATERIALS LIST

S.No	Part Name	Materials	Process Used	Quantity	Available Size	Required Dimensions
1.	Base	Mild Steel Sheet	Shearing, Drilling	1	2.0mm Thickness	400 mm x 160 mm x 2 mm
2	Chariot Side Plate	Mild Steel Sheet	Profile Cutting, Filing	2	1.0mm Thickness	300 mm x 160 mm x 2 mm
3	Fan	GI Sheet	Cutting, Bending	1	0.35mm Thickness	100 mm (Outer Dia)
4	Flag	GI Sheet, Mild Steel Rod	Cutting, Welding	1	0.35mm (Sheet), 4mm Dia (Rod)	150 mm X 44 mm
5	Front Tyres	Mild Steel Flat, Mild Steel Rod	Bending, Cutting, Welding	2	25mm x 3mm (Flat), 6mm Dia (Rod)	80 mm (Outer Dia.)
6	Seat	Mild Steel Sheet	Cutting, Bending, Welding	1	1.0mm Thickness	220 mm x 85 mm x 160 mm
7	Front Tyre Guard	Mild Steel Flat	Cutting, Bending	2	25mm x 3mm	150 mm arc, 30 mm width
8	Gears	Mild Steel Sheet	Cutting, Bending	2	2.0mm Thickness	130 mm (Outer Dia.)
9	Rear Tyre	Mild Steel Flat, Mild Steel Rod	Bending, Cutting, Welding	2	25mm x 5mm (Flat), 6mm Dia (Rod)	86 mm (outer dia.)
10	Axles	Mild Steel Rod	Cutting, Bending	2	10mm Dia	6 mm (dia.) x 100 mm

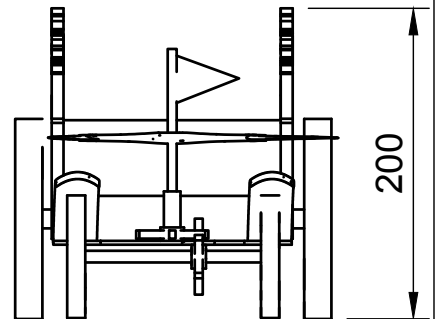
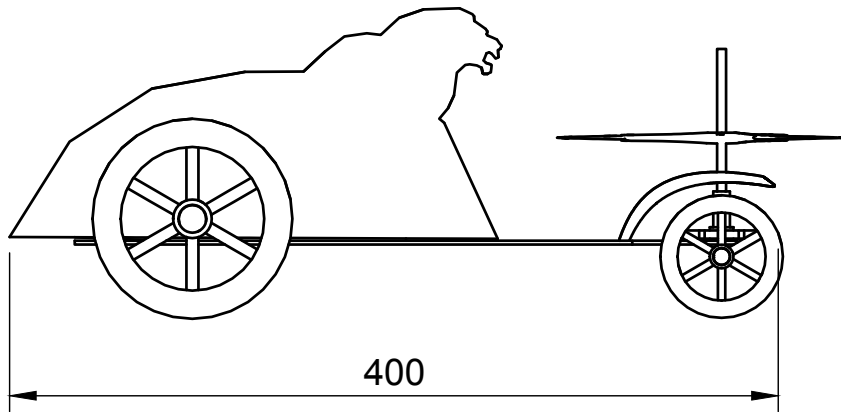
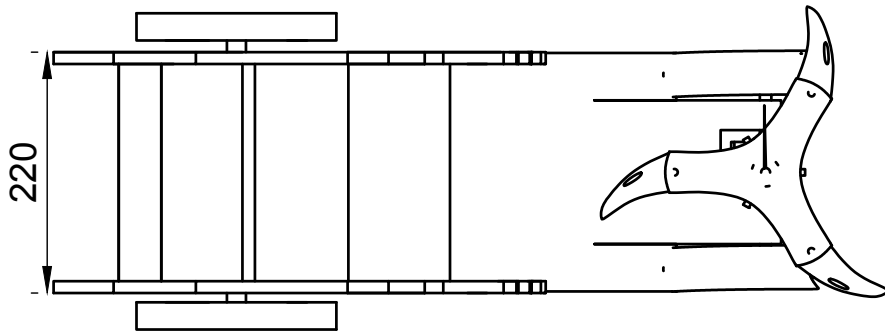
WORK DISTRIBUTION

<u>Name</u>	<u>P1</u>	<u>P2</u>	<u>P3</u>	<u>P4</u>	<u>P5</u>	<u>P6</u>
Lakshya Arukiya	Base	Front Wheels	Front Wheels	Gears	Gears	Assembly
Mannat Shrivastav	Base	Front Wheels	Front Wheels	Gears	Gears	Assembly
Manthan J	Side Tiger	Front Wheels	Front Wheel Guard	Seat	Seat	Assembly
Mayank Raj	Side Tiger	Rear Wheels	Front Wheel Guard	Seat	Seat	Assembly
Mayank Verma	Fan	Rear Wheels	Rear Wheels	Flag	Flag	Assembly
Meenakshi Meena	Fan	Rear Wheels	Rear Wheels	Flag	Flag	Assembly

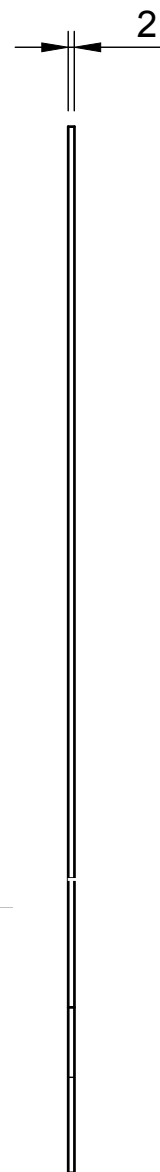
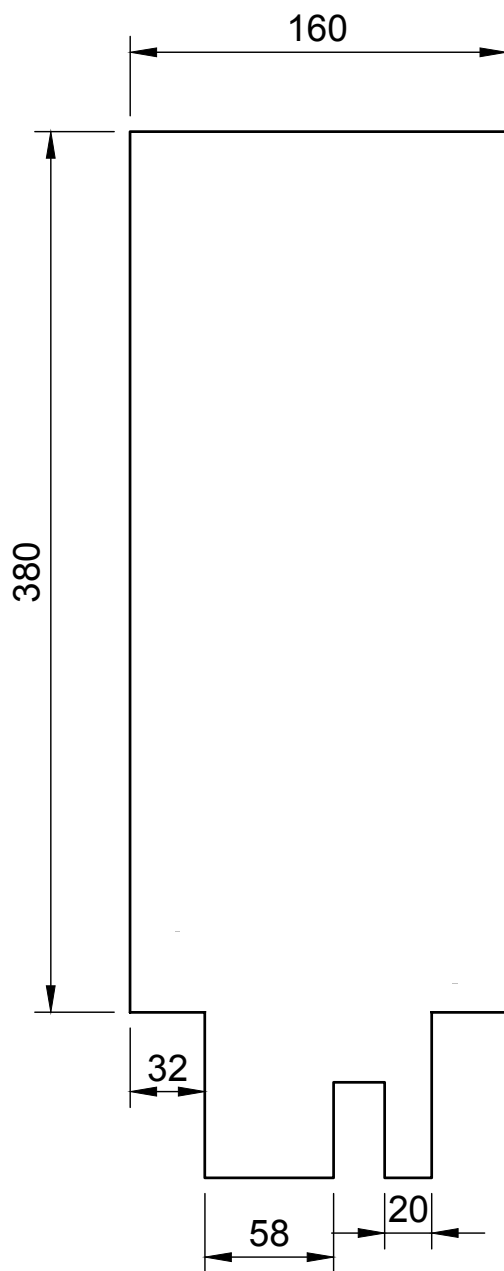


Part No.	Part Name
1	Base
2	Side Tiger (X2)
3	Fan
4	Flag
5	Front Tyres
6	Seat
7	Front Tyre Guard (X2)
8	Rear Tyres
9	Gears (X2)

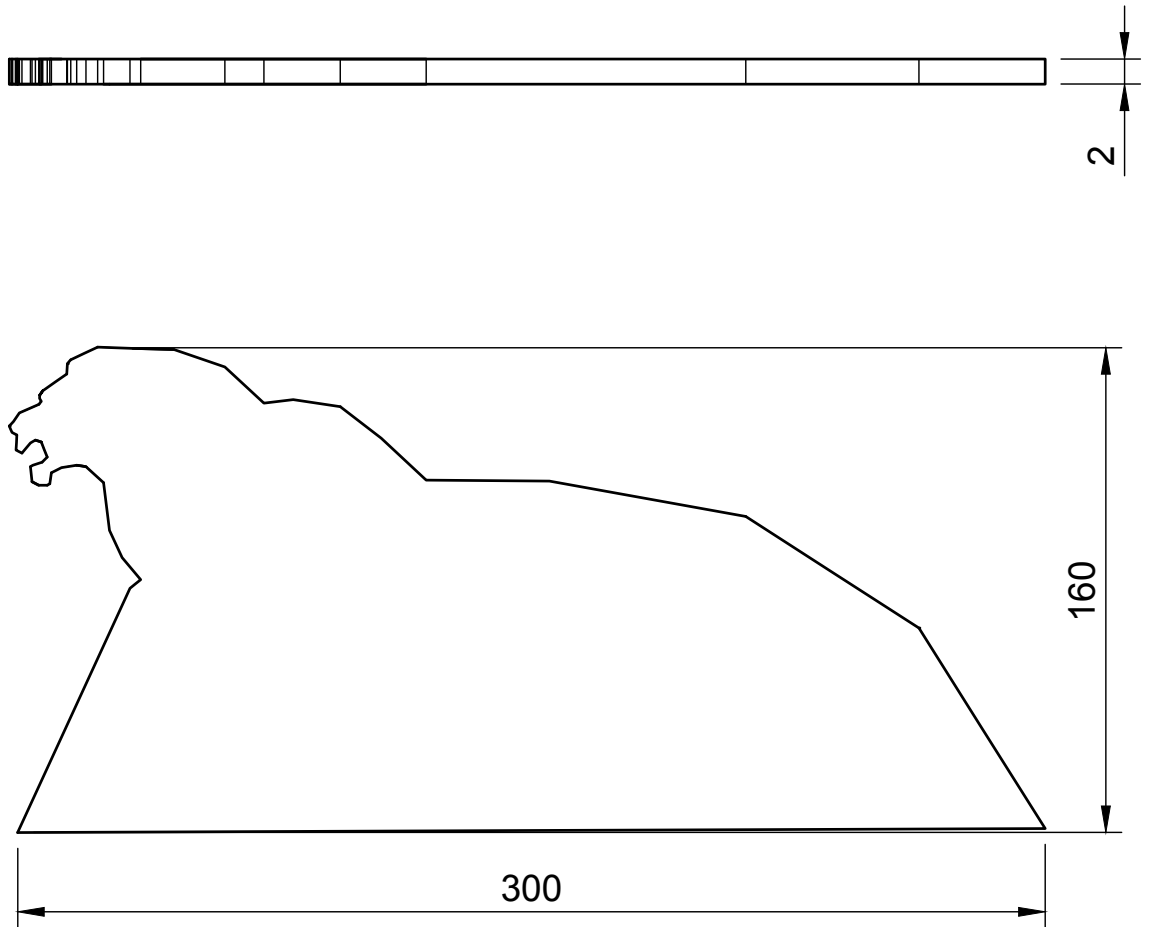
Dept.	Technical reference	Created by Lakshya Arukiya 01/09/25	Approved by		
		Document type	Document status		
		Title Isometric Assembly	DWG No.		
			Rev.	Date of issue	Sheet 1/11



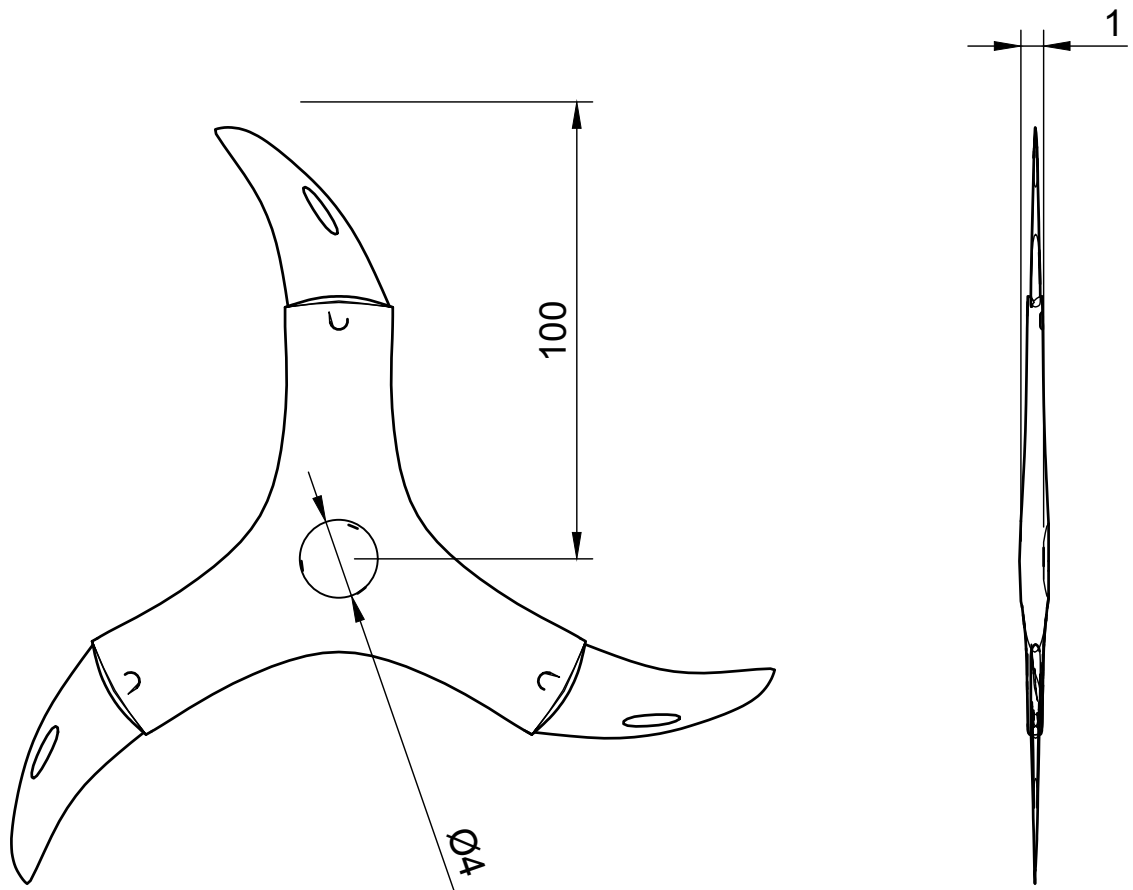
Dept.	Technical reference	Created by Lakshya Arukiya 01/09/25	Approved by		
		Document type	Document status		
		Title Orthographic Assembly	DWG No.		
			Rev.	Date of issue	Sheet 2/11



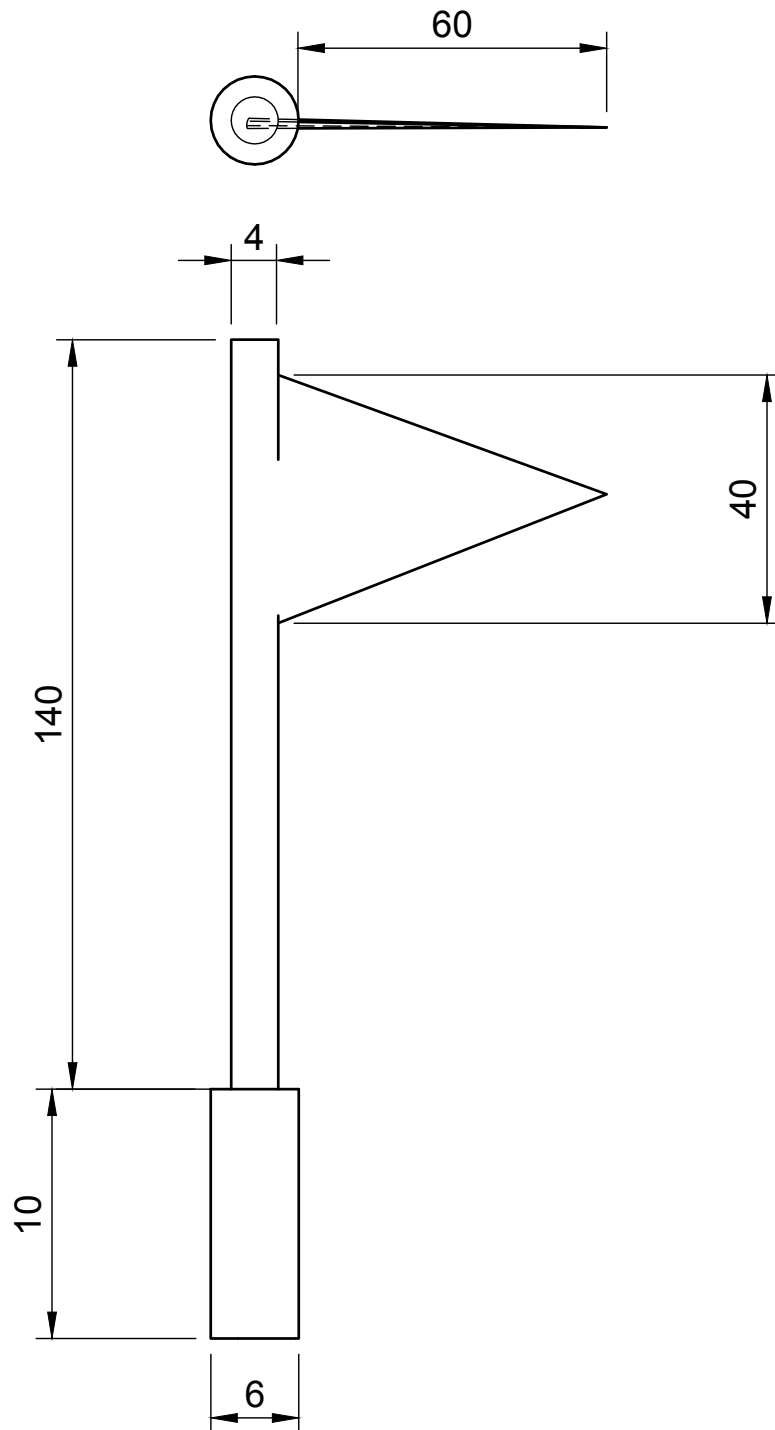
Dept.	Technical reference	Created by Mayank Raj	01/09/25			Approved by		
		Document type			Document status			
		Title Base			DWG No. Part 1			
					Rev.	Date of issue		Sheet 3/11



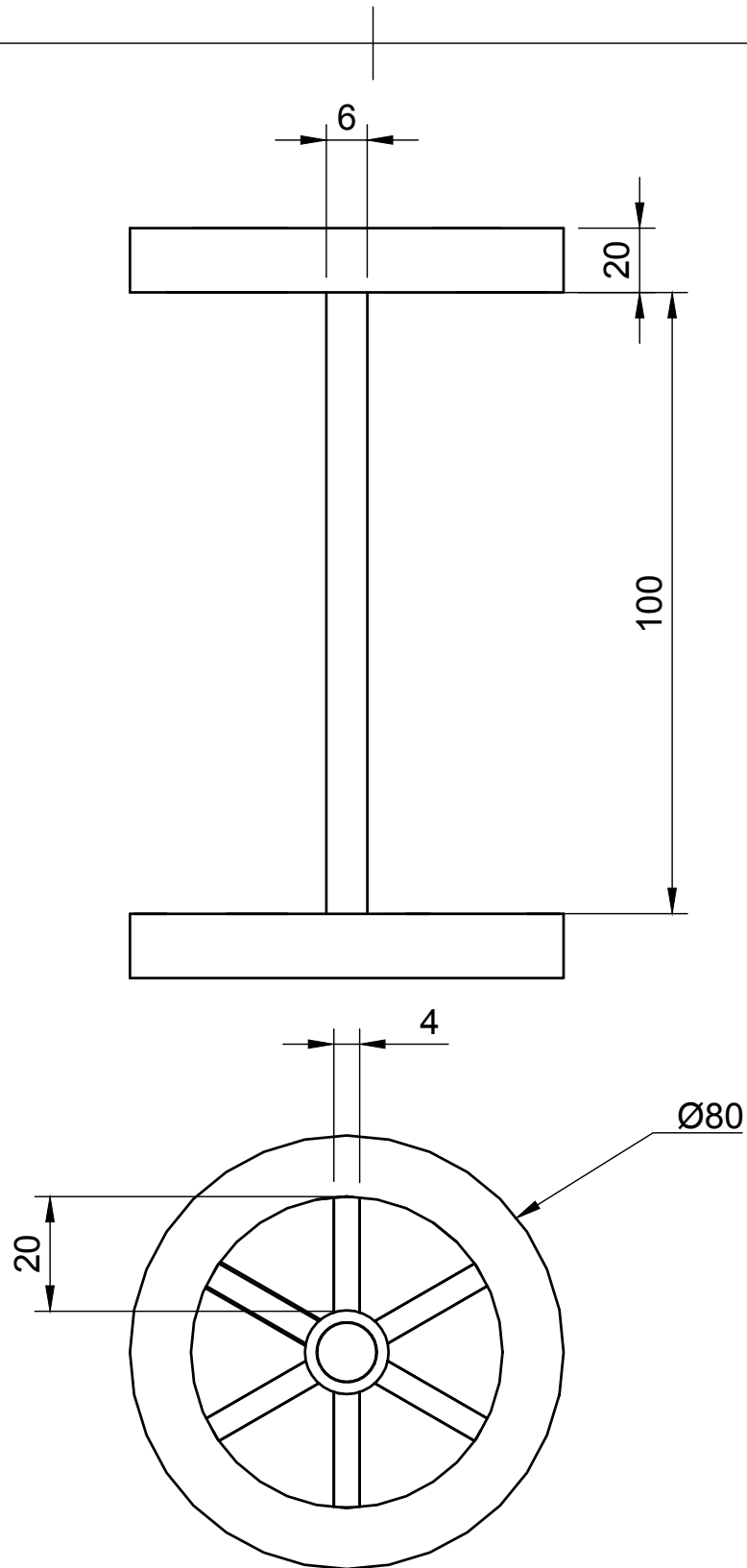
Dept.	Technical reference	Created by Manthan J	01/09/25			Approved by		
		Document type			Document status			
		Title Side Tiger (X2)			DWG No. Part 2			
					Rev.	Date of issue		Sheet 4/11



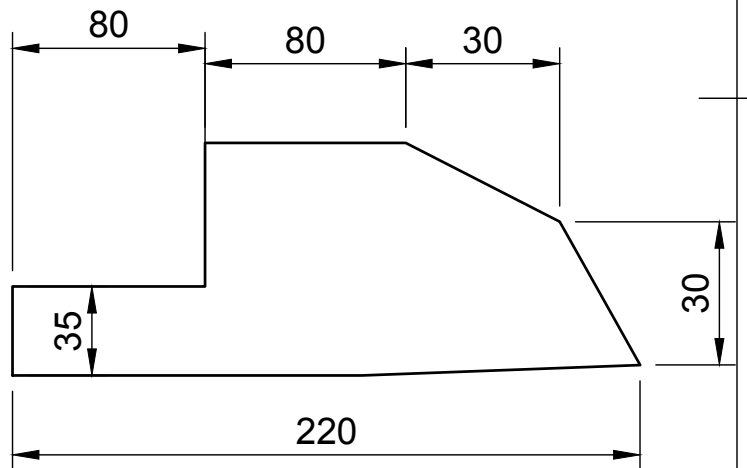
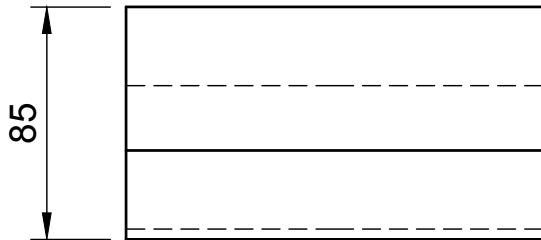
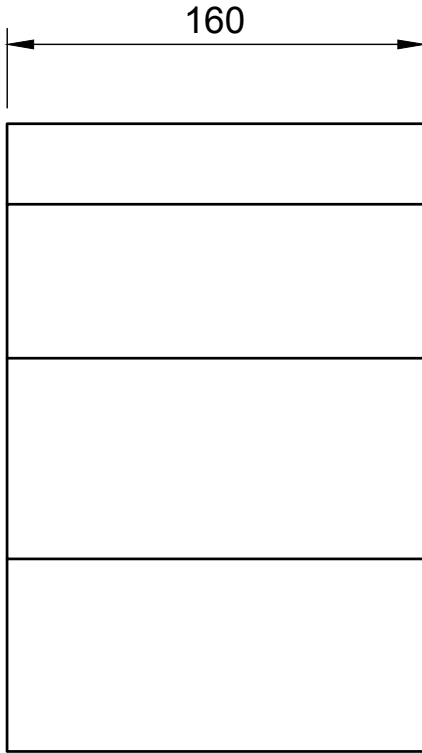
Dept.	Technical reference	Created by Mayank Verma 01/09/25	Approved by		
		Document type	Document status		
		Title Fan	DWG No. Part 3		
			Rev.	Date of issue	Sheet 5/11



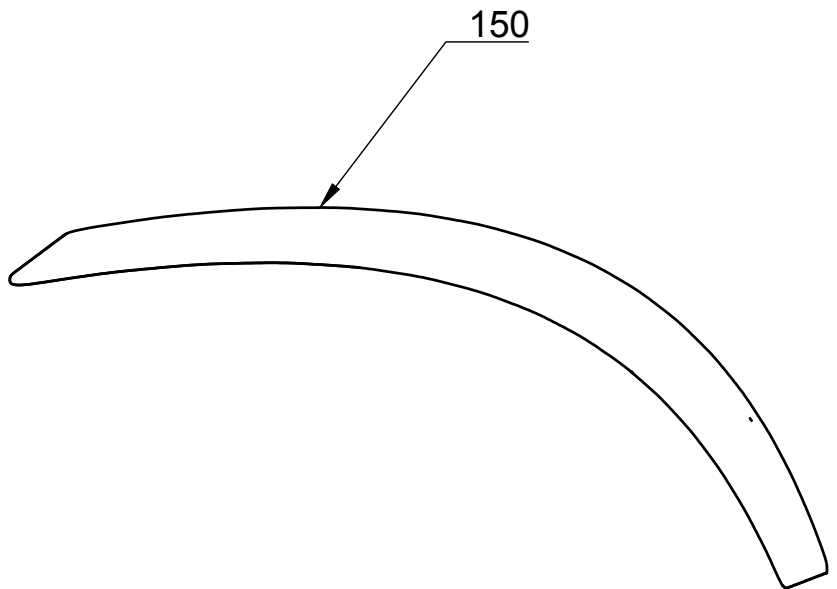
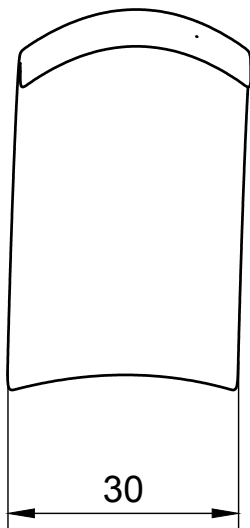
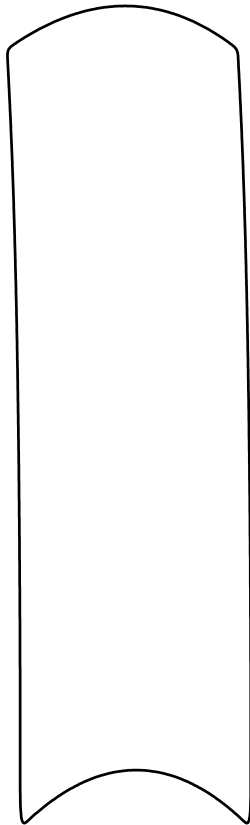
Dept.	Technical reference	Created by Lakshya Arukiya 01/09/25	Approved by		
		Document type	Document status		
		Title Flag	DWG No. Part 4		
			Rev.	Date of issue	Sheet 6/11



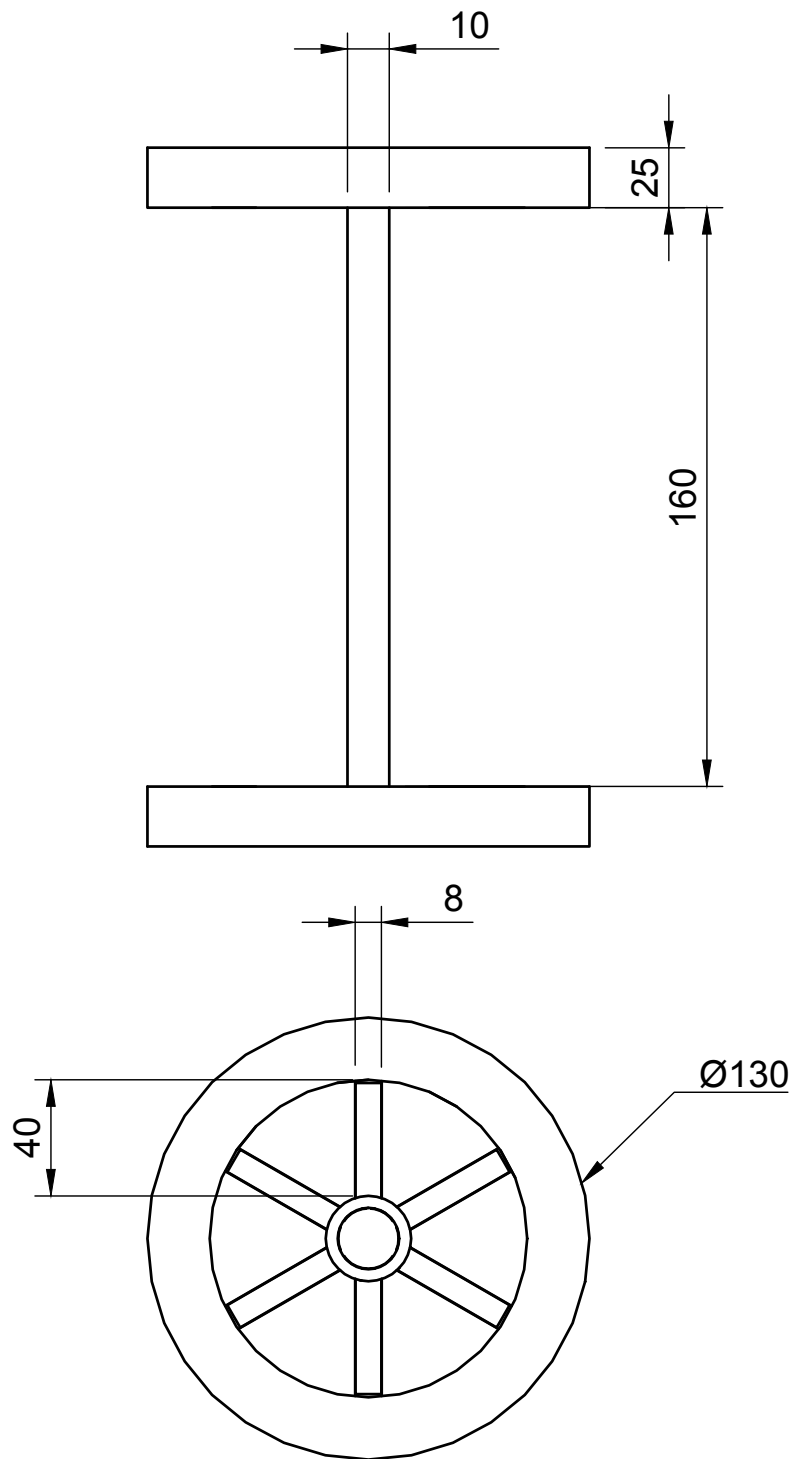
Dept.	Technical reference	Created by Mannat Shrivastav 01/09/25	Approved by		
		Document type	Document status		
		Title Front Tyres	DWG No. Part 5		
			Rev.	Date of issue	Sheet 7/11



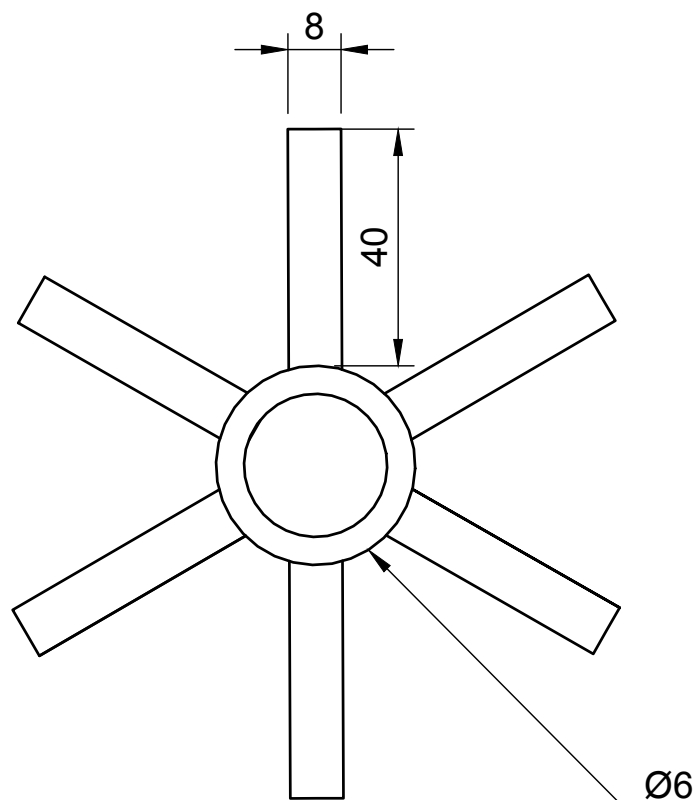
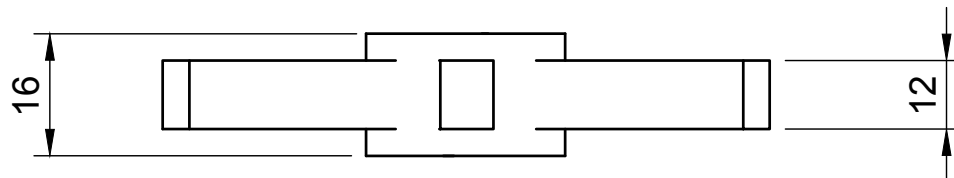
Dept.	Technical reference	Created by Lakshya Arukiya 01/09/25	Approved by		
		Document type	Document status		
		Title Seat	DWG No. Part 6		
			Rev.	Date of issue	Sheet 8/11



Dept.	Technical reference	Created by Meenakshi Meena 01/09/25	Approved by		
		Document type	Document status		
		Title Front Tyre Guard (X2)	DWG No. Part 7		
			Rev.	Date of issue	Sheet 9/11



Dept.	Technical reference	Created by Mayank Verma 01/09/25	Approved by		
		Document type	Document status		
		Title Rear Tyres	DWG No. Part 8		
			Rev.	Date of issue	Sheet 10/11



Dept.	Technical reference	Created by Mayank Raj	01/09/25		Approved by
		Document type	Document status		
		Title Gears (X2)	DWG No. Part 9		
			Rev.	Date of issue	Sheet 11/11